ANTOINE HENRI **BECQUEREL**

The French physicist Henri Becquerel was the first to discover natural radioactivity. In 1896, while investigating X-rays, he noticed that uranium emits radioactive particles.

Becquerel (1852–1908) was interested in phosphorescence (light-emitting substances). He wanted to see if there was any connection between phosphorescence and X-rays. He took a salt that contained uranium and was known to glow after exposure to light and put it on to some photographic plates, hoping that the uranium would absorb light and reemit it as X-rays. He left it in a dark drawer overnight, and the next day was amazed to find that the plates had been exposed. This proved that the uranium in the salt had emitted radiation. In recognition of his work, the unit of radioactivity is named after him: the becquerel.

Then, in 1906, tragedy struck: Pierre was knocked down by a carriage and killed. Left with two young daughters to raise, Curie also managed to take over her husband's professorship of physics at the Sorbonne. She continued with her research and finally isolated pure radium in 1910. For her exceptional work on the extraction and properties of this element, she won a second Nobel Prize in 1911, this time for chemistry.

Danger and value of radioactivity

Although Curie had coined the term "radioactivity" to describe how energy is released when atoms disintegrate into a different form, she was unaware of the dangers of handling radioactive substances and carried samples and stored them in her desk. The penetrating power of the rays given off by uranium and other radioactive substances makes them potentially dangerous, and Curie started to show signs of radiation sickness. In spite of this, scientists—including Curie—were starting to realize

"One never notices what **has been done**; one can only see **what remains to be done."**

Marie Curie, 1894